Transfer Learning Based Multilingual Neural Machine Translation for Low Resource Indic Languages

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***Abstract*—This research addresses the critical challenge of translating low resource Indic languages - specifically Bodo, Dogri, Konkani, and Maithili - into English using sequence-to-sequence models based on Transformer architecture. Here we are building a many-to-one neural machine Translation model. Conventional training methods prove inadequate due to the scarcity of high-quality parallel data sets for these languages. To navigate this issue, this study employs transfer learning and exploiting lexical similarity as an effective strategy to optimize a pretrained sequence-to-sequence model, mBART50, to accommodate the characteristics of these low resource languages. The training process capitalizes on similarities between the target languages and high resource languages, facilitating translation despite limited data availability. Through fine tuning a multilingual pretrained mbart-50 on low resource languages sentence pairs, we have achieved promising results on a IN22 gen benchmark data set with a blue score 29 on Bodo, 40 on Dogri, 26 on Konkani and 33 for Maithili. These findings showcase the potential for transfer learning to significantly enhance the accessibility and translation quality of low resource languages. Results of fine tuning showed efficient exploitation of parallel data from related language pairs. Future research should seek to refine these methods and expand the size of fine tuning data set, thereby extending the reach and benefits of machine translation.**

***Index Terms*—Transfer Learning, Fine tuning, Multilingual, Transformers, relatedness, NMT**

I. Introduction

In last decade Machine Translation have seen a huge progress. Introduction of sequence-to-sequence (Seq2Seq) Neural Machine Translation (Sutskever et al., 2014) and incorporating attention mechanism to these NMT (Bahdanau et al., 2015) have outperformed the Statistical Machine Translation (Kohen et al., 2009). These Seq2Seq NMT are made of Encoder-Decoder model based on Recurrent Neural Network. The encoder encodes the input or source sentence to a contextualized representation, while decoder decode this representation to the desired output or target sentence by maximizing the probability of target sentence given source sentence. The Transformers network-based encoder-decoder (Vaswani et al., 2017) led to a huge advancement in translation quality of NMT. Transformers are based on self-attention mechanism which are better at encoding a long sequences and removed the short comings of recurrent based sequence2sequence models by parallelizing the computations. In transformer based sequence2sequence models the encoder learns the more robust and contextualized representation of input sequence using self attention mechanism and the decoder uses this contextualized representation along with the target tokens to generate output in auto-regressive manner. These models require a huge data for the training process and have significantly improved the translation quality for high resource language due to the availability of monolingual and parallel data.

The dependency of NMT models on vast data creates a challenging scenario for less-resourced languages or low-resource settings. Low-resource languages, characterized by a limited volume of available parallel corpora, present an intriguing challenge. The lack of adequate data in these languages limits the performance of NMT systems, leading to poor translation quality. Consequently, users of these languages are often excluded from enjoying the benefits of this trans-formative technology. Moreover, the gap between well-resourced languages and low-resourced languages continue to widen, thereby exacerbating the digital divide and limiting information accessibility across different language communities. This issue primarily arises due to the model's inability to learn effective representations when trained with limited data.

Efforts have been made to overcome the problem of small corpora for low resource languages like backtranslation (Sennrich et al., 2016), data augmentation (Fadaee et al., 2017), self-learning algorithm (Zhang et al., 2016), semi-supervised method (Cheng et al., 2016), unsupervised MT (Artetxe et al., 2017: Lampele et al., 2017, 2018) and dual learning (He et al., 2016). Despite these efforts, transfer learning has shown a great potential to tackle this problem. Fine tuning a very small bilingual data set improved the translation accuracy via Transfer learning (Zoph et al., 2016). Instead of training a separate model for each language pair. A single model is trained on multiple languages by sharing parameters across all languages to perform multilingual NMT (Johnson et al.,

2017) using transfer learning and zero shot learning.

This study aims to address the challenge of insufficient data in low-resource Neural Machine Translation. It explores innovative strategies for enhancing the performance of NMT systems in low-resource settings, focusing on methods such as transfer learning, fine tuning, lexical transfer, syntactic transfer, language relatedness and multilingual model training. Our objective is to build a more equitable, low cost, and effective NMT system that can offer better translation of four low resource languages which are (i) Bodo, (ii) Dogri, (iii) Konkani and (iv) Maithili into English. These are very low resource languages spoken in Northern region of India where majority of people don't speak English. A very little study on the translation of these languages into English have been done where the translation quality was not feasible. (Gala et al., 2023) reported decent results on translation of these languages by training NMT from scratch which requires high-cost resources. Our study aims to build a multilingual NMT on these languages which is cost effective, simple and don't require training of model from scratch.

In this research, we present a (i) many-to-one multilingual NMT for low resource Indian languages (Bodo, Dogri, Konkani and Maithili) to English, (ii) show the results of transfer learning by fine tuning on low resource languages and compare the performance of our model with other publicly available or commercial neural machine translation models, (iii) how lexical and structural similarity is exploited in related languages.

II. Related Works

A. NMT based on Transformer Architecture

Transformer architecture based on self-attention mechanism (Vaswani et al., 2017) replaced the recurrence, and compute more robust and contextual representation of its input sequences. The original transformer based model had encoder and decoder. Each block in encoder has multi-head attention layers which are made up of many parallel self attention layers which pay attention to different semantics in a sequence, followed by feed forward network and some residual connection for better optimization. primary task of encoder is to encode the sequence of tokens into vectors while paying attention to the tokens in the vicinity of a particular token. Decoder is also made up of multi head attention and feed forward networks in a similar fashion to encoders. while training, decoder uses the output of encoder and shifted target inputs are fed via teacher forcing to remove the error propagation in auto-regressive process. While inference the decoder does not require inputs, it only requires the output of encoder if it is encoder-decoder based model. But these architectures were unidirectional in attention mechanism. This issue was resolved by BERT architecture which are deep bidirectional encoding (Devlin et al., 2019) of inputs using the same traditional transformer architecture. BERT can represent more robust contextualized representation from a text conditioning on both left and right context in layer. The architecture of transformer based encoder-decoder is shown in **Fig.1**

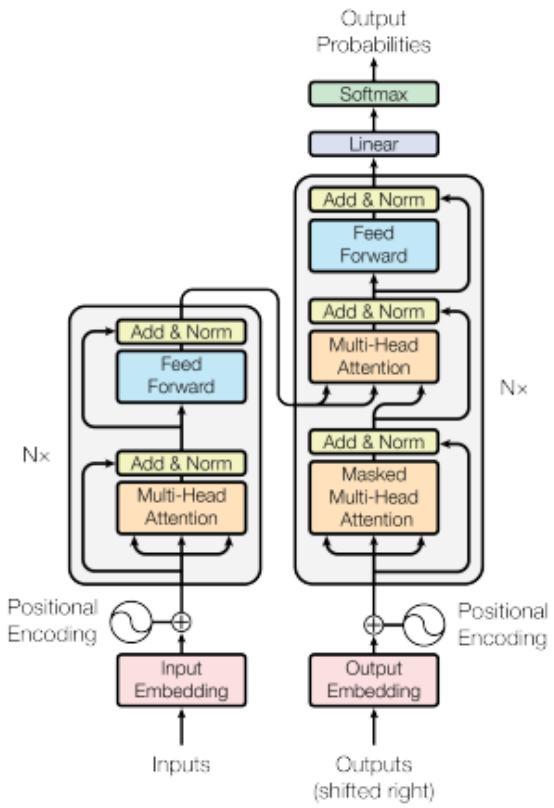


Fig. 1. Transformer model-architecture.(Vaswani et al., 2017)

Transformers are pretrained through mask language modelling (MLM) and next sentence prediction task (NSP) objective. Encoder-Decoder model based on pretrained Transformer architecture are used for Machine Translation. Sequence2sequence model can be made by combining a pretrained Bert encoder and using the Bert as a decoder with some extra condition's like auto-regresivness and masked attention mechanism. Fine-tuned encoder-decoder models based on BERT outperformed previous NMT models. Even we can also pretrained a sequence2sequence model from scratch like mbart, mT5 and MarianMT.

B. Multilingual NMT and Transfer Learning

Multilingual NMT based on a common encoder with separate decoder and attention mechanism (Dong et al., 2015) for each language was proposed. Multi-way, multilingual NMT (Firat et al., 2016) consisting of shared attention mechanism, although different encoder and decoders for multiple language pairs. NMT for low resource languages proposed by (Zoph et al., 2016) uses parent-child architecture, and that has been base line for majority of work. This model proposed the concept of Transfer Learning in which the parent model is pretrained on parallel data of high resource language pairs, then the child model is initialized with the parameters of parent model and then child model is fine-tuned on low resource language pairs. Multilingual NMT with single encoder and single decoder

with shared attention mechanism (Johnson et al., 2017) for multiple language pairs was introduced exploiting transfer learning and zero shot translations. The languages chosen in training of parent model affects the results of transfer learning on child model was studied (Dabre et al., 2018), which showed better results are obtained when parent model and child model language belongs to the same family. They actually make use of the shared vocabulary.

C. Language Relatedness and Unified Transliteration

India have 122 languages out of majority of these belongs to either Indo-European (spoken in Northern region) or Dravidian (spoken southern region) family. Under Indo-European, Indo-Aryan family is the dominant which compromises 21 languages. Our data set contains Bodo, Dogri, Maithili and Konkani as source and English as a target language. All the source language belongs to Indo-Aryan family and are written in Devanagari script. Languages belonging to a same family share vocabulary at word and character level, along with have same sentence structure which is Subject-Object-Verb (SOV) order. They are closely related to each other and have huge similarities. Common scripts for languages results in overlap and smaller sub word vocabulary (Ramesh et al., 2021). Multilingual NMT (Johnson et al., 2017) did not utilize the relatedness in low resource languages. To exploit the language relatedness and having shared vocabulary, Indic languages of Indo-Aryan family were transliterated to devnagri script and Dravidian family were transliterated to Abugida script (Vikrant et al., 2020; Jyotsana et al., 2021; Sourav et al., 2021). BPE algorithm were used for tokenization to have more similarities at vocabulary level. Low resource languages in our work are already have a common script and our model uses sentencepiece tokenizer which incorporates the BPE scheme.

III. Methodology

A. Data set

In our experiments, we use the sub data set of Bharat Parallel Corpus Collection (BPCC) by AI4BHARAT for fine tuning. This sub data set has sentence pairs of four languages i.e., Bodo, Dogri, Konkani, Maithili and their corresponding translation into English. This data set consists of sentences from five to six domains including daily use case, religion, sports, politics, economy, culture, news, entertainment etc. Data set contains sentence pairs which were manually translated by certified translators for Wikipedia and daily use cases and added back translated sentence pairs to make it more diverse. For testing we used IN22 gen test set (Gala et al., 2023) comprised sentence pairs from Wikipedia and Web sources subsets offer diverse content spanning of news, entertainment, geography, health, education, sports, tourism, industry, economy, legal and religion. IN22 gen is made by combining IN22-wiki and IN22-web. Second benchmark test set is IN22 Conv (Gala et al., 2023) which is also multi-domain and more diverse then IN22 gen. It includes sentence pairs from day-to-day conversation comprising of crime, disaster, politics, entertainment, geography, health, nature, science, sports, travel, politics. The details of parallel data for fine tuning and benchmark test data sets are summarized in **Table I**.

Data set Name	Language	No of Sent Pairs	Data set Name
Fine Tuning	Bodo	304,256	BPCC
	Dogri	270,223	
	Konkani	339,958	
	Maithili	327,120	
IN22 gen	Bodo	1024	IN22 gen test set
	Dogri	1024	
	Konkani	1024	
	Maithili	1024	
IN22 conv	Bodo	1503	IN22 conv test set
	Dogri	1503	
	Konkani	1503	
	Maithili	1503	

TABLE I
SUMMARIZATION OF FINE TUNING DATA SET AND BENCHMARK TEST DATA SET.

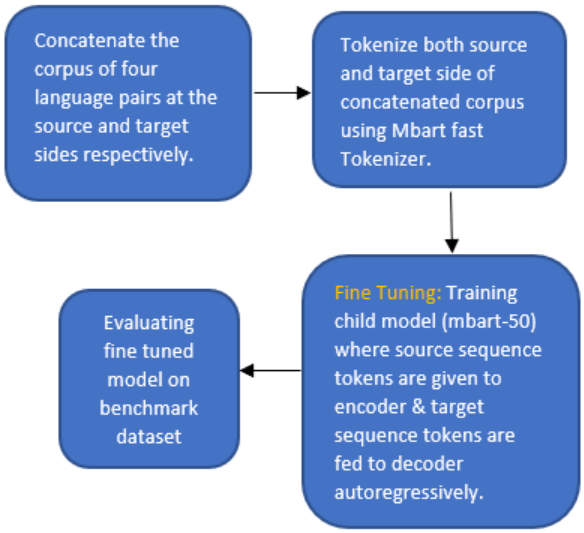


Fig. 2. Pipeline for fine tuning.

multirow

B. Data Preprocessing

For fine tuning process the sentence pairs of different languages are concatenated to create a training data set. Mbart fast tokenizer has been used for tokenization of sentence pairs. The tokenizer uses sentencepiece model (Kudo and Richardson, 2018) that includes 250, 000 sub word tokens. This tokenization supports fine-tuning on additional languages. No additional pre-processing like true casing or normalizing punctuation/characters has been applied.

C. Training Details/ Experiment Setup

We have used the pretrained multilingual mBart-large-50 (Tang et al., 2020) from HuggingFace. It is a multilingual Sequence-to-Sequence pretrained model. mBart-large-50 is extended version of original mBART ((Liu et al.,2020) to support 25 extra languages for multilingual machine translation.

During fine tuning the encoder is given input sentences to encode them into vector by exploiting the context of words in long sentences. The decoder is given these encoded vectors to their masked multi head attention layers and shifted target tokens are fed for the better optimization. The decoder uses both encoded inputs and target tokens to generate the expected sequence of tokens in auto-regressive manner while training. It is pretrained mainly for being fine-tuned on translation task. The architecture has 12 layers of encoder and 12 layers of decoder with model dimension of 1024 on 16 heads (680M parameters). each layer of encoder consists of multi-head attention layers followed by feed forward network along with layer normalization and residual connections. The code pipeline includes merging of translated sentence pairs from two different text files, tokenized using the model tokenizer. The tokenized source and target sentence pairs are mapped to embeddings, which are then fed to the encoder and decoder of the model respectively. The pipeline is shown by **Fig.2**. The maximum length of tokens for both source and target are kept 128. During fine tuning, the learning rate undergoes a linear increase, starting from 2e-5 and continuing for a specific number of initial steps. In this case, the warm-up phase consists of 500 steps. The model is fine-tuned for 10 epochs while applying a weight decay of 0.01. Finally, we evaluate the accuracy of our translation models using BLEU (papineni et al., 2002).

IV. Evaluation

A. Models compared

We have compared our fine-tuned model with publicly available models such as

- IndicTrans2 (Gala et al., 2023) built by AI4BHARAT team. They created Bharat Parallel Corpus Collection (BPCC) a largest available parallel corpus for Indic languages containing a total of 230M text pairs and used it to train encoder-decoder based multilingual transformer model with 434M parameters. This model can cater to 22 primary Indian languages.
- Microsoft Azure Translate. As of this document, Microsoft Azure Translate, a professional translation tool, offers translation services for 16 of the 22 scheduled Indic languages.
- Google Translate. Another commercial translation service, Google Translate, caters to 19 of the scheduled 22 Indic languages during the period this was documented.

B. Metric

BLEU (papineni et al., 2002) score is a popular metric used to evaluate the quality of machine generated translations or text generation outputs. For easier comparison in evaluation metric over different publicly available models and benchmark data sets, I have chosen BLEU score. It compares the candidate translation generated by model with reference translation produced by humans. The main task is to compute n-grams precision by counting the number of times n-gram of candidate appears in reference translation and then divide it by the length of candidate sentence. The meaning of n-grams is that how many words are being compared of candidate with reference, usually the value of n is taken from 1 to 4. The count of n-gram of candidate are clipped by the maximum reference count for that n-gram in reference translation to avoid the error caused by repetition of n-grams in the candidate sentence. In our results we have showed bleu score on the scale of 0-100.

$$p_n = \frac{\sum_{C \in \text{Candidates}} \sum_{\text{n-gram} \in C} \text{Count}_{\text{clip}}(\text{n-gram})}{\sum_{C_0 \in \text{Candidates}} \sum_{\text{n-gram}_0 \in C_0} \text{Count}_{\text{clip}}(\text{n-gram}_0)}$$

To calculate bleu scores over the entire test corpus, first we count the occurrences of all n-grams of candidate in reference sentences over the whole corpus and then divides it by the length of candidate corpus. Shorter sentences may have high bleu scores as we divide by the length of candidate, so to compensate this we add a brevity penalty for candidate with shorter length than the reference. The brevity penalty is also applied over the entire corpus instead of calculating it on each sentence and then averaging it. Finally, the bleu score is calculated by adding the brevity penalty to the modified n-gram precision score, where we take logarithmic of the computed geometric average of all n-grams (generally n varies from 1 to 4).

$$\log \text{BLEU} = \min\left(1 - \frac{r}{c}, 0\right) + \sum_{n=1}^{N} w_n \log p_n$$

For calculation of brevity score, c is the length of candidate translation and r is the effective length of corpus. Wn=1/N

V. Results

I report the results of many-to-one multilingual fine tuning on low resource languages (Bodo, Dogri, Konkani Maithili →English). **Table II** shows the difference in performance when the model was fine-tuned on each of the four languages separately in comparison with the multilingual fine tuning on IN22 gen benchmark.

Source language	bilingual FT	multilingual FT
Bodo	27	29
Dogri	37	40
Konkani	22	26
Maithili	30	33

TABLE II
BLEU SCORES COMPARISON OF BILINGUAL FINE TUNING AND MULTILINGUAL FINE TUNING ON IN22 GEN BENCHMARK DATA SET.

We have compared the BLEU score of our model with IndicTrans2 (Gala et al., 2023), google translator, Microsoft azure translator om IN22 gen data set in **Table III** and on IN22 conv dataset in **Table IV**.

VI. Discussion of Results

The BLEU scores in **Table II** highlights the improvement in the performance of multilingual fine tuned model over models which were fine tuned on each language translation pairs separately. We got an improvement of +2, +3, +4, +3 bleu

Source language	Our model	IndicTrans 2	Google	Azure
Bodo	29	40	-	-
Dogri	40	53	45	-
Konkani	26	35	33	25
Maithili	33	40	39	36

TABLE III
COMPARISON OF BLEU SCORES ON IN22 GEN OF OUR MODEL WITH VARIOUS PUBLICLY AVAILABLE MODELS.

Source language	Our model	IndicTrans2	Google	Azure
Bodo	29	35	-	-
Dogri	37	45	42	-
Konkani	24	29	29	23
Maithili	29	35	36	32

TABLE IV
COMPARISON OF BLEU SCORES ON IN22 CONV OF OUR MODEL WITH VARIOUS PUBLICLY AVAILABLE MODELS.

scores on Bodo, Dogri, Konkani and Maithili by multilingual fine tuning respectively. Fine tuning results clearly show that it is a case of Transfer learning as mbart was pretrained on Indian languages based on Indo-Aryan and Dravidian family which originated from the same Brahmi script, but it was not pretrained on any of these languages. Moreover, multilingual fine tuning with languages of same family (Indo-Aryan) and based on common script (Devnagri) significantly improves BLUE scores is observed due to better learning of language representation in shared encoder-decoder architecture and exploiting the lexical similarity and language relatedness. In **Table III** and **Table IV** we have compared the performance of our model on IN22 gen and IN22 conv benchmark with different model such IndicTrans2, google translator and Microsoft Azure translator. On comparing performance on IN22 gen with other models, our model lacks in translation quality by on average of 10 bleu score. The reason for that is that these models are pretrained on large scale i.e., more parameters and more languages than mbart, IndicTrans 2 model have been pretrained on nearly 3 million sentence pairs on each Bodo, Dogri, Konkani and Maithili comparing to 0.3 million sentence pairs in our fine tuning. Similarly, the google and azure translator have been pretrained on large architecture with more languages and with improved data sets.

In **Table IV** comparison of performance on IN22 conv benchmark which is more diverse and includes instances from daily conversation. One of the import points to note in **Table IV** is that performance of other models on IN22 conv data set compared to IN22 gen have decreased significantly around 5% for IndicTrans2, 3% for google but for our model it went down by nearly 2%. The gap in blue score of our model with other models decreased from 10 to 6. This shows that our fine-tuned model is more robust and generalized on daily conversations. our fine tuned with less computation score is good translator for daily conversation with people and for that you do not need to train a model from scratch. The computation cost for fine tuning our model compared to other three models is very low, which makes it more cost efficient and has a decent performance.

VII. CONCLUSION AND FUTURE WORKS

In this work, we explored different techniques such as Transfer learning, and Language Relatedness to improve the translation quality between Indian languages to English. Our results of multilingual fine-tuning were better than bilingual fine tuning. Multilingual fine tuning improved the translation quality by sharing sub word vocabulary and exploiting language relatedness. As we have seen that the model generalizes well on both In22 gen and IN22 conv benchmarks but still lacks in performance compared to other benchmark models. Although, we could not outperform the existing state of the art model such IndicTrans2 and google on IN22 benchmark data set but we reached closer to the performance of these large scale pretrained models just by fine tuning over a small data set. In future, we will try to improve results on benchmark by fine tuning it on more sentence pairs or trying powerful architectures. We can also try hierarchical transfer learning using high resource languages such as Hindi in first stage fine tuning and low resource languages in second stage fine tuning.

REFERENCES

1. Artetxe, M., 2017. Unsupervised Neural Machine Translation.

2. Bahdanau, D., 2015. Neural machine Translation by Jointly Learning to Align and Translate. ICLR.

3. Cheng, Y., 2016. Semisupervised learning for neural machine translation.

4. dabre, R., 2018. NICT'sParticipation in WAT2018: Approaches Using Multilingualism and Recurrently Stacked Layers.

5. Devlin, J., 2018. BERT Pretraining of Deep bidirectional Transformers for Language Understanding.

6. Doddapaneni, S., 2022. Towards Leaving No Indic Language Behind: Building Monolingual Corpora, Benchmark and Models for Indic Languages.

7. Dong, D., 2015. Multi-Task Learning for Multiple Language Translation.

8. Fadaee, M., 2017. Data Augmentation for Low-Resource Neural Machine Translation.

9. Firat, O., 2016. Multi-Way, Multilingual Neural Machine Translation with a Shared Attention Mechanism.

10. Gala, J., 2023. IndicTrans2 Towards high Quality and Accessible Machine Translation models for All 22 scheduled indian languages.

11. Goyal, V., 2020. Efficient Neural Machine Translation for Low-Resource Languages via Exploiting Related Languages.

12. Johnson, M., 2017. Google's Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation.

13. Khatri, J., 2021. Language Relatedness and Lexical Closeness can help Improve Multilingual NMT: IITBombay@MultiIndicNMT WAT2021.

14. Koehn, P., 2009. Statistical machine Translation.

15. Kudo, T., 2018. SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing.

16. Kumar, S., 2021. Multilingual Multi-Domain NMT for

Indian Languages.
17. Lampele, 2017. Unsupervised machine translation using monolingual corpora only.
18. Liu, Y., 2020. Multilingual Denoising Pre-training for Neural Machine Translation.
19. Papineni, K., 2002. Bleu: a Method for Automatic Evaluation of Machine Translation.
20. Ramesh, A., 2021. Comparing Statistical and Neural Machine Translation Performance on Hindi-To-Tamil and English-To-Tamil.
21. Sennrich, R., 2015. Neural Machine Translation of rare Words with Subword units.
22. Sutskever, I., 2015. Sequence to Sequence Learning with Neural networks. p. 9.
23. Tang, Y., 2020. Multilingual Translationwith Extensible Multilingual Pretraining and Finetuning.
24. Vaswani, A., 2017. Attention Is All You Need.
25. Xia, Y., 2016. Dual learning for Machine Translation.
26. Zhang, J., 2016. Exploiting Source-side Monolingual Data in Neural Machine Translation.
27. Zoph, B., 2016. Transfer Learning for Low-Resource Neural Machine Translation.

Appendix

Here I have compared some translation generated by the model with actual translation for each of the languages.

Example No:	Bodo to English
1	**Sentence to be Translated:** ब्रितिश खौरां बिलाइ द गार्जियननि मथै ड्यूश बैंका बिखौ मावफुंनो बाहायनाय 1200 कम्पेनिफोरनि गेजेराव थामनि से बाहागोखौ दबथादोंमोन। **Translation by Model:** According to British newspaper The Guardian, Deutsche Bank controlled one-third of the 1200 companies it used to execute it. **Actual Translation:** British newspaper The Guardian suggested Deutsche Bank controlled roughly a third of the 1200 shell companies used to accomplish this.
2	**Sentence to be Translated:** गासै बुहुमाव अजदफोर, गोबां दायगिरिफोरखौ साजा होनाय जादोंमोन, आरो आइसलैन्ड आरो पाकिस्तानननि सरखारफोरनि दैदेनगिरिफोर बिबानगिरि एंगारदोंमोन। **Translation by Model:** There were lawsuits, multiple convicts were convicted all over the world, and leaders of the governments of Iceland and Pakistan resigned. **Actual Translation:** There were protests worldwide, several criminal prosecutions, and the leaders of the governments of Iceland and Pakistan both resigned.

Example No:	Dogri to English
1	**Sentence to be Translated:** ब्रिटिश अख़बार द गार्जियन दे मताबक ड्यूश बैंक ने इसगी पूरा करने आस्तै इस्तेमाल कीती जाने आह्ली 1200 शेल कंपनियें बिच्चा इक तिहाई हिस्से गी मुट्टे तौरा पर नियंत्रत कीता। **Translation by Model:** According to British newspaper The Guardian the Deutsche Bank purchased a third of the 1,200 shell companies it used to complete it. **Actual Translation:** British newspaper The Guardian suggested Deutsche Bank controlled roughly a third of the 1200 shell companies used to accomplish this.

2	**Sentence to be Translated:** दुनिया भर च बरोध प्रदर्शन होए, केईं अपराधक मुकदमे दर्ज होए ते आइसलैंड ते पाकिस्तान दियें सरकारें दे केइयें नेताएं अस्तीफा देई दित्ता। **Translation by Model:** There were protests all over the world, many criminal trials were lodged, and many leaders of the governments of Iceland and Pakistan were deserted. **Actual Translation:** There were protests worldwide, several criminal prosecutions, and the leaders of the governments of Iceland and Pakistan both resigned.

Example No:	Konkani to English
1	**Sentence to be Translated:** हे सुवातेर पांच गगनचुंबी इमारतींचे बांदकाम चालू आसा, तांचे मद्दीं येरादारी केंद्र आनी यादस्तीक पार्क आसा. **Translation by Model:** Five skyscrapers are under construction on the site, with a transport hub and a memorial park in their vicinity. **Actual Translation:** Construction is ongoing for five new skyscrapers at the site, with a transportation center and memorial park in the middle.
2	**Sentence to be Translated:** दरेक शो ल्हान भुरगीं तांच्या ग्रंथालयांत वतकच खंयचीं पुस्तकां सोदचीं तेविशींय शिफारशी करता. **Translation by Model:** Each show also recommends which books to look for when young children visit their library. **Actual Translation:** Each show would also provide recommendations for books that children should look for when they went to their library.

Example No:	Maithili to English
1	**Sentence to be Translated:** ◌्रिटिश अखबार द गार्जियन सुझाव देलथि की ड्यूश बैंक एहि काज कें संपन्न करबाक लेल उपयोग कएल गेल 1200 शेल कंपनी सभ मे सँ लगभग एक तिहाई कें नियंत्रित केलथि। **Translation by Model:** British newspaper The Guardian suggested Deutsche Bank controlled roughly a third of the 1200 shell companies used to accomplish this. **Actual Translation:** Construction is ongoing for five new skyscrapers at the site, with a transportation center and memorial park in the middle.
	Sentence to be Translated: दुनिया भरि मे विरोध प्रदर्शन भेल, कएक आपराधिक मोकदमा दाखिल कएल गेल आ आइसलैंड आओर पाकिस्तान, दुनू देशक सरकारक नेता त्यागपत्र देलथि।

2	**Translation by Model:** Protests erupted around the world, with multiple criminal prosecutions being filed, and leaders of governments in both Iceland and Pakistan resigning. **Actual Translation:** There were protests worldwide, several criminal prosecutions, and the leaders of the governments of Iceland and Pakistan both resigned.